

Overview of Curse of dimensionality

Subject: Curse of dimensionality

$$V_d(r) = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)} r^d$$

1.

The curse of dimensionality refers to various phenomena that arise when analyzing and organizing data in high-dimensional spaces that do not occur in low-dimensional settings such as the three-dimensional physical space of everyday experience. The expression was coined by Richard E. Bellman when considering problems in dynamic programming. The curse generally refers to issues that arise when the number of datapoints is small (in a suitably defined sense) relative to the intrinsic dimension of the data. Dimensionally cursed phenomena occur in domains such as numerical analysis, sampling, combinatorics, machine learning, data mining and databases. The common theme of these problems is that when the dimensionality increases, the volume of the space increases so fast that the available data become sparse. In order to obtain a reliable result, the amount of data needed often grows exponentially with the dimensionality. Also, organizing and searching data often relies on detecting areas where objects form groups with similar properties; in high dimensional data, however, all objects appear to be sparse and dissimilar in many ways, which prevents common data organization strategies from being efficient.

2.

Therefore, the squared distance from the origin, $r^2 = \sum_i x_i^2$ has the average value $d/3$ and variance $4d/45$. For large d , distribution of r^2/d is close to the normal distribution with the mean $1/3$ and the standard deviation $2/\sqrt{45d}$ according to the central limit theorem. Thus, when uniformly generating points in high dimensions, both the "middle" of the hypercube, and the corners are empty, and all the volume is concentrated near the surface of a sphere of "intermediate" radius $\sqrt{d/3}$. This also helps to understand the chi-squared distribution. Indeed, the (non-central) chi-squared distribution associated to a random point in the interval $[-1, 1]$ is the same as the distribution of the length-squared of a random point in the d -cube. By the law of large numbers, this distribution concentrates itself in a narrow band around d times the standard deviation squared (σ^2) of the original derivation. This illuminates the chi-squared distribution and also illustrates that most of the volume of the d -cube concentrates near the boundary of a sphere of radius $\sigma \sqrt{d}$. A further development of this phenomenon is as follows. Any fixed distribution on the real numbers induces a product distribution on points in $\mathbb{R}^d$. For any fixed n , it turns out that the difference between the minimum and the maximum distance between a random reference point Q and a list of n random data points $P_1, \dots, P_n$ become indiscernible compared to the minimum distance:

3.

k-nearest neighbor classification Another effect of high dimensionality on distance functions concerns k-nearest neighbor (k-NN) graphs constructed from a data set using a distance function. As the dimension increases, the indegree distribution of the k-NN digraph becomes skewed with a peak on the right because of the emergence of a disproportionate number of hubs, that is, data-points that appear in many more k-NN lists of other data-points than the average. This phenomenon can have a considerable impact on various techniques for classification (including the k-NN classifier), semi-supervised learning, and clustering, and it also affects information retrieval.

4.

Blessing of dimensionality Despite the expected "curse of dimensionality" difficulties, common-sense heuristics based on the most straightforward methods "can yield results which are almost surely optimal" for high-dimensional problems. The term "blessing of dimensionality" was introduced in the late 1990s. Donoho in his "Millennium manifesto" explained why he thinks the "blessing of dimensionality" will form a basis of future data mining. The effects of the blessing of dimensionality were discovered in many applications and found their foundation in the concentration of measure phenomena. One example of the blessing of dimensionality phenomenon is linear separability of a random point from a large finite random set with high probability even if this set is exponentially large: the number of elements in this random set can grow exponentially with dimension. Moreover, this linear functional can be selected in the form of the simplest linear Fisher discriminant. This separability theorem was proven for a wide class of probability distributions: general uniformly log-concave distributions, product distributions in a cube and many other families (reviewed recently in). "The blessing of dimensionality and the curse of dimensionality are two sides of the same coin." For example, the typical property of essentially high-dimensional probability distributions in a high-dimensional space is: the squared distance of random points to a selected point is, with high probability, close to the average (or median) squared distance. This property significantly simplifies the expected geometry of data and indexing of high-dimensional data (blessing), but, at the same time, it makes the similarity search in high dimensions difficult and even useless (curse). Zimek et al. noted that while the typical formalizations of the curse of dimensionality affect i.i.d. data, having data that is separated in each attribute becomes easier even in high dimensions, and argued that the signal-to-noise ratio matters: data becomes easier with each attribute that adds signal, and harder with attributes that only add noise (irrelevant error) to the data. In particular for unsupervised data analysis this effect is known as swamping.

5.

V h y p e r s p h e r e V h y p e r c u b e $= \pi^{d/2} / (2^d \Gamma(d/2)) \rightarrow 0$ as $d \rightarrow \infty$. Furthermore, the distance between the center and the corners is $r\sqrt{d}$, which increases without bound for fixed r . In this sense when points are uniformly generated in a high-dimensional hypercube, almost all points are much farther than r units away from the center. In high dimensions, the volume of the d -dimensional unit hypercube (with coordinates of the vertices ± 1) is concentrated near a sphere with the radius $\sqrt{d}/\sqrt{3}$ for large dimension d . Indeed, for each coordinate x_i the average value of x_i^2 in the cube is